

CO3-AT3-Assignment

View Only

← Back

QUESTIONS

Q1 
Banking Transaction System –
Exception Handling
10 marks

Q2 
Hospital Patient Registration –
User-Defined Exception
10 marks

Q3 
E-Commerce Order Processing
– Built-in Exceptions
10 marks

Q4 
Hospital Emergency System –
Thread Priority
10 marks

Q5 
Online Food Delivery – Multiple
Threads and Execution Order
10 marks

5/5 answered

Q1 Banking Transaction System – Exception Handling

10 marks

Develop a Java program for an ATM withdrawal system. The program accepts the account balance and withdrawal amount.

The program must:

- Accept the current account balance and withdrawal amount.
- Throw a user-defined `InsufficientBalanceException` when the withdrawal amount exceeds the balance.
- Handle withdrawal amounts that are zero or negative.
- Handle invalid numeric input using a suitable built-in exception.
- Display a meaningful transaction result.
- Use try-catch-finally so that the transaction completion message is always displayed.

Sample Test Cases:

Input / Scenario Expected Result Purpose

10000 3000 Normal case

10000 12000 Boundary / exception case

10000 -500 Invalid input

abc 3000 Invalid input

Your Code

```
1 import java.util.*;
2 public class Main{
3     static class InsufficientBalanceException extends Exception{
4         public InsufficientBalanceException(String message){
5             super(message);
6         }
7     }
8     public static void main(String[] args){
9         Scanner sc = new Scanner(System.in);
10        try{
11            double balance = Double.parseDouble(sc.next());
12            double withdrawl=Double.parseDouble(sc.next());
13            if(withdrawl<=0){
14                System.out.println("Invalid withdrawl amount");
15            }
16            else if(withdrawl>balance){
17
18                throw new InsufficientBalanceException("Insuffcient balance");
19            }
20            else{
21                balance = balance - withdrawl;
22            }
23            System.out.println("Withdrawl sucessful");
24            System.out.println("Remaining balance: "+balance);
25
26        }
27        catch(InsufficientBalanceException e){
28            System.out.println(e.getMessage());
29        }
30        catch(NumberFormatException e){
31            System.out.println("Invalid numeric input");
32        }
33        finally{
34            System.out.println("Transaction completed");
35        }
36        sc.close();
37    }
}
```

Input

10000
3000

Output

Withdrawl sucessful
Remaining balance: 7000.0
Transaction completed



✔ Submitted